# Operimus - Marketing

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| Project Abstract: | This project involves the development of a Marketing Automation Platform that integrates campaign management, lead tracking, and content creation. It allows users to automate workflows, schedule multi-channel campaigns, and personalize templates. The system provides real-time analytics to improve efficiency, collaboration, and data-driven decision making. |

# Index

Operimus - Marketing ..... 1
Index ..... 2
1 Introduction ..... 3
2 Product concept ..... 3
2.1 Vision statement ..... 3
2.2 Personas and Scenarios ..... 3
2.3 Scenarios ..... 5
2.4 EPICS ..... 6
2.5 Product requirements (User stories) ..... 10
2.5 Score ..... 33
2.6 Requirements ..... 35
3 Architecture notebook ..... 37
3.1 Architetural view ..... 37
Fg1. - MAP architecture ..... 37
3.2 Module interactions ..... 37

# 1 Introduction

This project, developed within the Software Engineering (ES) course, focuses on the design of a **Marketing Automation Platform (MAP)**. The platform aims to streamline and automate marketing activities such as email campaigns, social media management, and lead scoring, enabling teams to concentrate on strategy and content creation. Its core functionalities include **lead management, automated workflows, email and social media tools, landing pages, reporting, and CRM integration**. Beyond technical specification, the project highlights how information systems engineering can improve marketing efficiency, decision-making, and business performance.

# 2 Product concept

## 2.1 Vision statement

The **Marketing Automation Platform (MAP)** is designed to help organizations **automate, personalize, and optimize marketing activities** while addressing the challenge of maintaining efficiency and customer engagement at scale. It will be used for tasks such as automated email and social media campaigns, lead management and segmentation, analytics, and CRM integration, enabling marketing teams to reduce manual work and improve decision-making.

## 2.2 Personas and Scenarios

**Marketing Manager:**

**Persona – Artur Borges**

Artur is a 35-year-old Marketing Manager who thrives in fast-paced, results-driven environments. He has over a decade of experience in marketing strategy and is highly skilled at juggling multiple campaigns across email, social media, and digital advertising. Artur is deeply analytical but also creative; he enjoys crafting campaigns that not only perform well but also tell a compelling story to his audience. His day-to-day involves coordinating with content creators, designers, and sales teams to ensure messaging is consistent, relevant, and aligned with business objectives.

**Motivation:** Artur uses the platform to streamline marketing processes, automate lead nurturing, deliver personalized campaigns, and track performance metrics, enabling him to optimize ROI and make data-driven decisions efficiently.

**Content Marketer:**

**Persona – Rafael Santos**

Rafael is a 28-year-old Content Marketer responsible for creating engaging content across multiple channels, including email campaigns and social media. He collaborates closely with designers, marketers, and sales teams to ensure that all content aligns with brand messaging and campaign objectives. Rafael is detail-oriented and creative, always experimenting with new formats, visuals, and messaging to capture audience attention. He relies on analytics to understand which types of content resonate most, refine messaging, and maximize engagement.

**Motivation:** Rafael uses the platform to streamline content creation and scheduling, test email templates, manage content libraries efficiently, and analyze audience engagement, allowing him to deliver relevant, high-performing content consistently.

**Sales Representative:**

**Persona – Felipe Fernandes**

Felipe is a 32-year-old Sales Representative focused on converting qualified leads into customers. He works closely with marketing to follow up on prospects who show interest in products or services. Felipe is highly relationship-driven and values having detailed insights about each leads behavior to tailor his sales approach. He monitors leads' engagement with marketing content, including emails opened, links clicked, and pages visited, to prioritize follow-ups and personalize conversations. By combining intuition with data, Felipe aims to increase conversion rates and strengthen client relationships.

**Motivation:** Felipe uses the platform to receive real-time alerts about engaged leads, access detailed interaction histories, and personalize his sales pitch, enabling him to prioritize efforts and convert leads more effectively.

**Marketing Analyst:**

**Persona - Mariana Lopes**

Mariana is a 30-year-old Marketing Analyst responsible for analyzing marketing performance and deriving actionable insights. She works closely with marketing managers, content creators, and sales teams to ensure campaigns are data-driven and optimized for results. Mariana is highly analytical, detail-oriented, and skilled at translating complex datasets into clear, actionable recommendations. She uses data to identify trends, test different campaign strategies, and measure ROI, ensuring that marketing efforts are effective and continuously improving.

**Motivation:** Mariana uses the platform to create custom dashboards, generate detailed reports, and run A/B tests, enabling her to measure campaign performance accurately, optimize strategies, and support data-driven decision-making.

## 2.3 Scenarios

### Scenario - Artur Borges (Marketing Manager)

Artur opens the platform to launch a new product campaign. He selects the “Create Workflow” option and designs a multi-channel journey: first an email announcing the product, then a follow-up email to those who clicked the link, and finally a social media post for broader reach. He segments the audience, ensuring tailored messages for each group. After setting the automation live, Artur checks real-time performance metrics such as open rates and conversions to optimize the campaign.

### Scenario – Rafael Santos (Content Marketer)

Rafael logs onto the platform to prepare next week’s content. He drafts several email templates for an upcoming promotion and tests different subject lines. He then schedules social media posts across social networks. From the analytics dashboard, he notices that video posts perform better than static images, so he decides to adapt future campaigns with more video content.

### Scenario – Felipe Fernandes (Sales Representative)

Filipe receives a real-time notification on the platform that a lead has opened three emails and clicked a product link. He opens the lead profile and reviews the history: the lead visited the pricing page twice and downloaded a brochure. Based on this engagement, Filipe prioritizes contacting the lead immediately. When he calls, he personalizes the conversation by highlighting the product features the lead showed interest in, increasing the chance of closing the sale.

### Scenario – Mariana Lopes (Marketing Analyst)

Mariana accesses the platform to evaluate the latest campaign’s results. She creates a custom dashboard showing KPIs like lead-to-customer conversion rate, ROI, and engagement levels across channels. To refine the strategy, she sets up an A/B test comparing two different email designs. After analyzing the results, Mariana discovers that personalized subject lines improved open rates by 25%. She documents these insights in a report and shares recommendations with Artur and Rafael to guide future campaigns.

## 2.4 EPICS

### Epic 1: Authentication & Security

**Description:** This epic focuses on providing secure access to the system by enabling users to log in and log out safely. The goal is to ensure that only authorized users can access their accounts while keeping data protected.

**User Stories included:**

1. User Login
2. User Logout

### Epic 2: Lead Management

**Description:** This epic involves capturing, organizing, segmenting, and scoring leads. The purpose is to help marketing managers identify high-value prospects and prioritize engagement effectively.

**User Stories included:**

1. Group leads into segments
2. Score leads based on actions

### Epic 3: Marketing Automation & Workflows

**Description:** This epic addresses the creation and management of automated workflows with triggers and actions. It allows marketing managers to nurture leads, personalize communication, and reduce manual intervention.

**User Stories included:**

1. Build automated marketing workflows
2. Configure Triggers and Actions in Workflows
3. Test and preview workflows
4. Edit existing workflows

### Epic 4: Campaign Management

**Description:** This epic covers the setup and execution of marketing campaigns across multiple channels. It enables managers to link workflows, manage materials, personalize campaigns, and track campaign performance.

**User Stories included:**

1. Create and manage campaigns
2. Manage campaign materials
3. Run campaigns across multiple channels
4. Track campaign results
5. Export reports
6. Track and update campaign status
7. Create campaigns from reusable workflow templates

## Epic 5: Email Template Management

**Description:** This epic focuses on the creation, editing, personalization, and management of email templates. It ensures that content marketers can deliver professional, personalized, and reusable templates for campaigns.

**User Stories included:**

1. Create and save email templates
2. Edit email templates
3. Personalize email templates
4. Delete email templates
5. Define A/B tests

## Epic 6: Landing Page Management

**Description:** This epic enables content marketers to design, edit, and manage landing pages that support campaigns and lead capture. The goal is to provide an easy way to create campaign-specific web pages without developer involvement.

**User Stories included:**

1. Create landing pages
2. Edit landing pages

3. Delete landing pages

## Epic 7: Form Management

**Description:** This epic involves creating, editing, and managing forms that capture user data and integrate it into campaigns. It ensures smooth lead collection and supports marketing workflows.

**User Stories included:**

1. Create forms
2. Edit forms
3. Delete forms

## Epic 8: Social Media Publishing

**Description:** This epic addresses the scheduling and management of social media posts. It enables content marketers to plan campaigns in advance and maintain consistent communication across channels.

**User Stories included:**

1. Schedule posts for social media
2. View scheduled posts

## Epic 9: Lead Management

**Description:** This epic covers the monitoring and management of individual leads. It provides sales representatives with real-time insights into lead activity, scoring, and engagement history, enabling them to prioritize outreach, personalize conversations, and keep CRM records up to date.

**User Stories included:**

1. Receive notifications about leads
2. View Leads email history
3. Leads webpage visited
4. Leads forms submitted
5. Full leads timeline interactions
6. Check lead score
7. Suggested talking points based on lead activity

## Epic 10: CRM Data Synchronization

**Description:** This epic ensures seamless synchronization between the marketing automation platform and external CRM systems. It allows sales representatives to update lead status, sync lead data, and avoid duplicates, ensuring that customer information is always accurate, up-to-date, and consistent across platforms.

**User Stories included:**

1. Sync lead information with CRM
2. Update lead status in the CRM from our platform

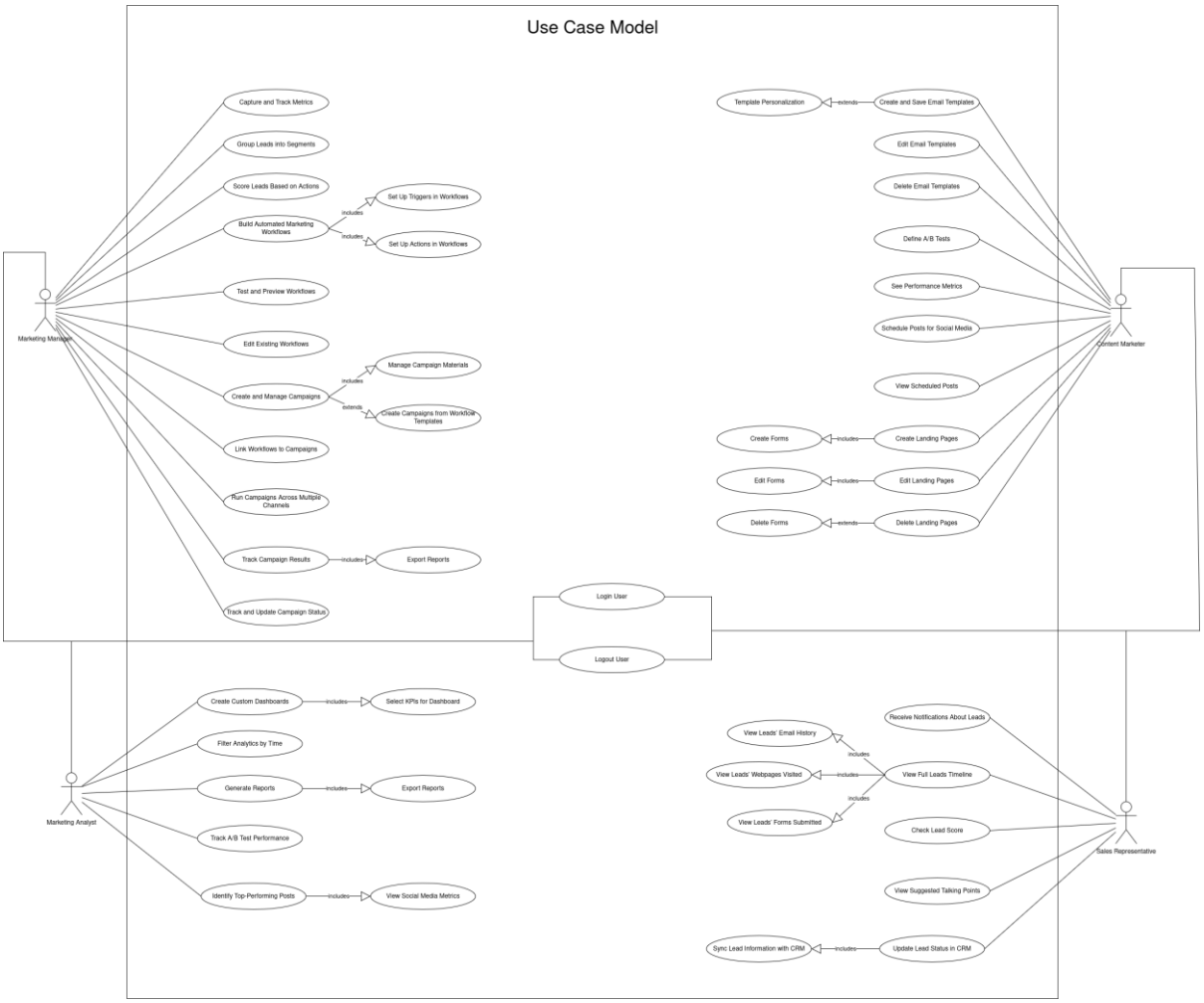
## Epic 11: Analytics & Dashboards

**Description:** This epic focuses on the creation, customization, and interpretation of marketing analytics. It allows marketing analysts to design dashboards, filter data, track campaign performance, and generate reports that reveal actionable insights and measure ROI across multiple channels.

**User Stories included:**

1. Create custom dashboards
2. Select which data appears in the dashboard
3. Generate reports
4. Export reports
5. Filter analytics by time
6. Track A/B test performance
7. View social media metrics
8. Identify top-performing posts
9. See performance metrics

## 2.5 Product requirements (User stories)



**All Users:**

**User Story 1.1:** User login

As a User, I want to log in to the system with my credentials, so that I can access my personalized account.

- **Scenario**: Successful login with valid credentials

  Given the user is on the login page

  When the user enters valid credentials

  And the user clicks the login button

  Then the user should be redirected to the Campaigns page

  And the username and role should be visible in the navbar

**User Story 1.2:** User logout

As a User, I want to log out of the system, so that my account remains secure when I'm done using it.

- **Scenario:** Successful logout from any page

  Given the user is on the login page

  When the user enters valid credentials

  And the user clicks the login button

  And the user clicks the logout option

  Then the session should end and redirect to the Keycloak login page

**Artur Borges – Marketing Manager**

**User Story 2.1:** Group leads into segments

As a Marketing Manager, I want to group leads into segments (e.g., by industry or engagement), so that I can target them better

- **Scenario:** Segmenting leads dynamically

  Given segmentation rules are defined by the Marketing Manager

  When leads match these criteria

  Then the system groups them automatically into segments

  And updates occur efficiently and reflect changes in real time

**User Story 2.2:** Score leads based on actions

As a Marketing Manager, I want to score leads based on their actions, so that I can focus on the best prospects.

- **Scenario:** Automatically scoring leads

  Given leads perform actions such as opening emails or visiting pages

  When the system evaluates their behavior

  Then their scores are updated correctly

  And qualified leads are flagged for follow-up without delay

**User Story 3.1:** Build automated marketing workflows

As a Marketing Manager, I want to build automated marketing workflows, so that leads are nurtured without manual work.

- **Scenario:** Successful creation and configuration of a new workflow

  Given the manager is on the login page

  When the manager enters valid credentials

  And the manager clicks the login button

  Then the manager should be redirected to the Campaigns page

  And the manager selects the campaign named "Winter Clearance"

  Then the manager should be redirected to the Campaign Details page

And the manager clicks the Create New Workflow button

Then the manager should be redirected to the Create Workflow page

And the manager fills in the workflow details with a name and description

And the manager confirms the workflow creation

Then the workflow should be successfully created

And the manager should be redirected back to the Campaign Details page

And the manager clicks the View Workflow button

Then the manager should be redirected to the Workflow Builder page

And the manager adds nodes to the workflow

And the manager connects nodes in the workflow

And the manager clicks the Save button

Then the workflow should be saved successfully and visible in the builder

**User Story 3.2:** Configure Triggers and Actions in Workflows

As a Marketing Manager, I want to configure **triggers** (e.g., email opened) and **actions** (e.g., send a follow-up email, notify sales),
**so that** the workflow can automatically react to lead behavior and execute the appropriate marketing steps.

- **Scenario:** Reacting to lead behavior with triggers and executing actions

  Given a workflow contains a defined trigger (e.g., “Email Opened”)

  When the trigger condition is met by a lead

  Then the workflow proceeds automatically

  And all associated actions (e.g., send follow-up email, notify sales) are executed in sequence

  And the workflow logs results for tracking

**User Story 3.3:** Test and preview workflows

As a Marketing Manager, I want to test and preview workflows before launch, so that I can ensure the automation runs correctly.

- **Scenario:** Safely previewing workflows

  Given a workflow is fully configured

  When the Marketing Manager runs a test

  Then the workflow simulates all steps without changing live data

  And the results are clearly displayed for review

**User Story 3.4:** Edit existing workflows

As a Marketing Manager, I want to edit existing workflows, so that I can update triggers, actions, or steps without rebuilding them from scratch

- **Scenario:** Successful edit of an existing workflow

  Given the manager is on the login page

  When the manager enters valid credentials

  And the manager clicks the login button

  Then the manager should be redirected to the Campaigns page

  And the manager selects the campaign named "Winter Clearance"

  Then the manager should be redirected to the Campaign Details page

  And the manager clicks the View Workflow button

  Then the manager should be redirected to the Workflow Builder page

  When the manager adds a Form block to the workflow

  And the manager connects nodes in the workflow

  And the manager saves the edited workflow

Then the workflow should be updated successfully and visible in the builder

**User Story 4.1:** Create and manage campaigns

As a Marketing Manager, I want to create and manage campaigns with a name, description, deadline, and materials, so that I can organize and run structured marketing initiative

- **Scenario:** Creating and updating campaigns

  Given the manager is on the login page

  When the manager enters valid credentials

  And the manager clicks the login button

  Then the manager should be redirected to the Campaigns page

  And the manager clicks the Create Campaign button

  And the manager fills in the campaign details with a name, description, and deadline

  And the manager saves the new campaign

  Then the campaign should be created and visible in the campaign list

**User Story 4.2:** Manage campaign materials

As a Marketing Manager, I want to manage campaign materials (emails, landing pages, forms), so that I can organize a campaign.

- **Scenario:** Managing campaign content

  Given campaign materials such as images or messages exist

  When the Marketing Manager edits or replaces them

  Then updates are stored correctly

  And changes do not affect unrelated campaigns

**User Story 4.3:** Run campaigns across multiple channels

As a Marketing Manager, I want to run campaigns across multiple channels, so that I can reach leads wherever they are

- **Scenario:** Executing multi-channel campaigns

  Given campaign content is prepared for email, social, and web

  When the campaign is launched

  Then messages are distributed reliably across all selected channels

  And delivery results are recorded for tracking

**User Story 4.4:** Track campaign results

As a Marketing Manager, I want to track campaign results (e.g., open and click rates), so that I can measure success

- **Scenario:** Viewing campaign performance metrics

  Given a campaign has collected engagement data

  When the Marketing Manager opens the analytics dashboard

  Then accurate metrics are displayed per campaign

  And charts reflect real-time results

**User Story 4.5:** Export reports

As a Marketing Manager, I want to export reports, so they can be talked in meetings.

- **Scenario:** Exporting performance reports

  Given a filtered report is displayed on screen

  When the Marketing Manager exports it

  Then a corresponding PDF is generated promptly

And the file is ready for download or sharing

**User Story 4.6:** Track and update campaign status

As a Marketing Manager, I want to track and update the status of a campaign (Pending, Active, Finished), so that I can quickly see which campaigns are scheduled, running, or completed.

- **Scenario:** Managing campaign status

  Given a campaign has a defined status

  When the Marketing Manager updates it (e.g., from Draft to Active)

  Then the change is saved correctly

  And other campaigns remain unaffected

**User Story 4.7:** Create campaigns from reusable workflow templates

As a Marketing Manager, I want to create campaigns from reusable workflow templates, so that I can quickly launch campaigns with proven automation flows while still customizing them for specific goals.

- **Scenario:** Using workflow templates for campaigns

  Given workflow templates are available

  When the Marketing Manager creates a campaign from one

  Then a new independent workflow is generated

  And modifications to it do not alter the original template

**Rafael Santos – Content Marketer**

**User Story 5.1:** Create and save Email Templates

As a Content Marketer, I want to create and save a new email template so that I can use it in current and future campaigns without starting from scratch.

- ***Scenario:*** *Creating reusable email templates*

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  When the content creator navigates to the Email Templates section

  Then the Email Templates page should be displayed

  When the content creator clicks the Create Email Template button

  Then the Create Email Template page should open

  And the content creator fills in the template name, description, and subject

  And the content creator previews the template

  And the content creator saves the new email template

  Then a success message should appear confirming the template creation

  And the content creator returns to the Email Templates page

  Then the new email template should appear in the templates list

**User Story 5.2:** Edit email templates

As a Content Marketer, I want to edit an existing email template so that I can update content or layout.

- **Scenario:** Updating existing email templates

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

When the content creator navigates to the Email Templates section

Then the Email Templates page should be displayed

And the content creator opens the template named "Test Email Template" for editing

And the content creator edits the name and description to "Email Test"

And the content creator previews the template

And the content creator saves the updated email template

Then a success message should appear confirming the template update

And the updated template card should display the new name "Email Test"

**User Story 5.3:** Personalize email templates

As a Content Marketer, I want to add personalization (e.g., customer name) to an email so that communication feels more relevant.

- **Scenario:** Adding personalization to emails

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  When the content creator navigates to the Email Templates section

  Then the Email Templates page should be displayed

  And the content creator opens the template named "Email Test" for editing

  And the content creator edits the subject to "Campaign {CompanyName}"

  And the content creator previews the template

  And the content creator saves the updated email template

Then a success message should appear confirming the template update

And the updated template should contain the subject "Campaign Operimus"

**User Story 5.4:** Delete Email Templates

As a Content Marketer, I want to delete an existing email template so that I can remove outdated or unused templates from the library.

- **Scenario:** Removing unused templates

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  When the content creator navigates to the Email Templates section

  Then the Email Templates page should be displayed

  And the content creator deletes the email template named "Email Test"

  Then a success message should appear confirming the template deletion

  And the template named "Email Test" should no longer appear in the list

**User Story 5.5:** Define A/B tests

As a Content Marketer, I want to set A/B versions of different elements of an email (e.g., subject lines, body content), so that I can identify which version performs better and optimize campaign results.

- **Scenario:** Setting up A/B email tests

  Given the Content Marketer creates multiple email versions

  When A/B testing is configured

  Then the system distributes sends between versions automatically

And reports the winning version based on open and click rates

And warns if the contact list is too small

And results are shown in the marketing dashboard

**User Story 6.1:** Create and previe Landing Pages

As a Content Marketer, I want to create a landing page with a drag-and-drop editor and customizable forms, so that I can capture leads without coding.

- **Scenario:** Building a new landing page

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  When the content marketer navigates to the Landing Pages section

  Then the Landing Pages page should be displayed

  When the content marketer clicks the Create Landing Page button

  Then the Create Landing Page form should open

  And the content marketer fills in the landing page name and description

  And the content marketer clicks the Preview Landing Page button

  Then a preview modal should appear showing the landing page preview

  And the content marketer saves the new landing page

  Then a success message should confirm the landing page creation

  And the content marketer returns to the Landing Pages page

  Then the new landing page should appear in the list of landing pages

**User Story 6.2:** Edit Landing Pages

As a Content Marketer, I want to edit an existing landing page so that I can update content, layout, or forms without recreating the page.

- **Scenario:** Updating existing landing pages

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  When the content marketer navigates to the Landing Pages section

  Then the Landing Pages page should be displayed

  Given the content marketer is on the Landing Pages list with at least one item

  When the content marketer selects an existing landing page to edit

  Then the landing page editor should open

  When the content marketer modifies the name and description

  And saves the edited landing page

  Then a confirmation message should appear before overwriting

  And the landing page should be successfully updated

**User Story 6.3:** Delete Landing Pages

As a Content Marketer, I want to delete a landing page so that I can remove outdated or unused pages from the system.

- **Scenario:** Removing outdated landing pages

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

Then the content creator should be redirected to the Campaigns page

When the content marketer navigates to the Landing Pages section

Then the Landing Pages page should be displayed

Given the content marketer is on the Landing Pages list with at least one item

When the content marketer selects it for deletion

Then the system asks for confirmation

And confirms the deletion

Then a success message should appear confirming the landing page deletion

And the deleted landing page should no longer appear in the list

**User Story 7.1:** Create Forms

As a Content Marketer, I want to create a form with customizable fields so that I can collect the right information from leads.

- **Scenario:** Building new lead capture forms

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  Given the Content Marketer is on the form builder page

  When they fill in the form name as "Registration Form" and the description as "Form to register new users"

  And they click on "Save Form"

  Then the form is saved successfully

  And the form name and description appear in the form list

**User Story 7.2:** Edit Forms

As a Content Marketer, I want to edit an existing form so that I can adjust questions without recreating it.

- **Scenario:** Updating existing forms

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  Given a saved form exists with name "Registration Form" and description "Form to register new users"

  When the Content Marketer edits the form name to "Updated Registration Form" and description to "Updated form for user registration"

  And they confirm saving the form

  Then the form is saved successfully after editing

  And the updated form name and description appear in the form list

**User Story 7.3:** Delete Forms

As a Content Marketer, I want to delete a form so that I can remove outdated or unused forms from the system.

- **Scenario:** Deleting unused forms

  Given the content creator is on the login page

  When the content creator enters valid credentials

  And the content creator clicks the login button

  Then the content creator should be redirected to the Campaigns page

  Given a form exists in the library with name "Updated Registration Form"

When the Content Marketer chooses to delete it

Then the system confirms the action before removal

And shows a success message once removed

And the form is no longer in the list

**User Story 8.1:** Schedule post for social media

As a Content Marketer, I want to schedule a post on a social media platform so that I can publish at the optimal time.

- **Scenario:** Scheduling social media posts

  Given the Content Marketer selects a social platform and account

  When they compose and schedule a post with date and time

  Then the system confirms successful scheduling

  And posts appear in the visual content calendar

  And permission errors are shown clearly if access is missing

  And scheduling integrates seamlessly with the content calendar

**User Story 8.2:** View scheduled posts

As a Content Marketer, I want to view all scheduled posts in a list so that I can manage content consistency.

- **Scenario:** Viewing and managing scheduled posts

  Given workflow templates are available

  When the Marketing Manager creates a campaign from one

  Then a new independent workflow is generated

  And modifications to it do not alter the original template

**User Story 11.9: See performance metrics**

As a Content Marketer, I want to access the performance metrics already available for my content (such as opens, clicks, and reach) so that I can quickly understand what resonates with the audience.

- **Scenario:** Viewing content performance metrics

  Given performance data is available for email or campaign content

  When the Content Marketer opens the analytics view

  Then key metrics such as opens, clicks, and reach are displayed

  And can be filtered by campaign and timeframe

  And "No data available" appears if no results exist

  And metrics remain consistent with ROI reports

**Felipe Fernandes – Sales Representative**

**User Story 9.1: Receive notifications about leads**

As a Sales Representative, I want to receive real-time notifications when a lead becomes hot so that I can follow up immediately.

- **Scenario:** Receiving real-time hot lead notifications

  Given a lead score increases to "hot"

  When the system detects this change

  Then a real-time notification is sent to the Sales Representative

  And it integrates with the lead scoring and CRM systems

  And the notification is clear, non-intrusive, and immediately visible

**User Story 9.2:** View Leads email history

As a Sales Representative, I want to view a lead email open history so that I know which messages they engaged with.

- **Scenario:** Checking a lead email engagement

  Given a Sales Representative opens a lead profile

When viewing their email history

Then the system displays a list of emails with open status and timestamps

And shows “No emails opened yet” if none are available

And data comes from the email marketing and CRM modules

And the view includes readable dates and filters

**User Story 9.3:** Leads webpages Visited

As a Sales Representative, I want to view the web pages visited by a lead so that I can understand their interests.

- **Scenario:** Viewing a lead website activity

  Given a Sales Representative opens a lead profile

  When viewing their web activity

  Then the system lists all visited pages with timestamps

  And shows “No data available” if tracking is disabled

  And activity data is pulled from the tracking and lead management modules

  And the view presents interests clearly in a list or heatmap

**User Story 9.4:** Leads forms submitted

As a Sales Representative, I want to see the forms a lead has submitted so that I know what information they have already provided.

- **Scenario:** Checking lead form submissions

  Given a Sales Representative opens a lead profile

  When viewing submitted forms

  Then the system displays all forms with submission dates and field values

  And hides sensitive fields when necessary

  And shows “No submissions recorded” if none exist

And uses data from the landing page and forms modules

**User Story 9.5:** Full leads timeline interactions

As a Sales Representative, I want to see a timeline of all lead interactions so that I have a clear overview before contacting them.

- **Scenario:** Viewing a complete interaction timeline

  Given a Sales Representative opens a lead profile

  When viewing their interaction timeline

  Then the system consolidates emails, pages, forms, notifications, and scores

  And displays events chronologically with date and type filters

  And shows an empty timeline message if no data exists

  And pulls all records from connected modules and CRM

**User Story 9.6:** Check Lead Score

As a Sales Representative, I want to check the lead score so that I can prioritize my outreach.

- **Scenario:** Viewing the lead score and priority level

  Given a Sales Representative opens a lead profile

  When checking their score

  Then the current score and details are displayed in real time

  And visual indicators (bar or color coding) show cold, warm, or hot levels

  And "Lead score unavailable" appears if data is missing

**User Story 9.7**: Suggested talking points based on the lead activity

As a Sales Representative, I want to view suggested talking points based on the lead activity so that I can personalize my sales pitch.

- **Scenario:** Receiving AI-generated talking points

  Given a Sales Representative opens a lead profile

  When suggestions are generated from lead activity

  Then the system displays short, clear, and actionable talking points

  And shows "No suggestions available" if data is insufficient

  And pulls insights from emails, forms, and page visits for personalization

**User Story 10.1:** Sync lead information with CRM

As a Sales Representative, I want the lead information to sync with the CRM so that I can manage opportunities without duplicate data.

- **Scenario:** Keeping lead data synchronized

  Given the CRM integration is active

  When the Sales Representative updates lead information

  Then the changes sync automatically in both directions

  And errors are shown if the CRM is offline, with retries handled gracefully

And clear feedback confirms when synchronization succeeds

**User Story 10.2:** Update leads status in the CRM from our platform

As a Sales Representative, I want to update leads status in the CRM directly from the platform so that my workflow is faster.

- **Scenario:** Changing leads CRM status directly

  Given a Sales Representative has permission to modify leads

  When they update the status (e.g., Contacted, Opportunity)

  Then the CRM reflects the new status automatically

  And "Access denied" appears if permissions are missing

  And a confirmation message verifies the update

  And duplication or manual re-entry is prevented

**Mariana Lopes – Marketing Analyst**

**User Story 11.1:** Create custom dashboards

As a Marketing Analyst, I want to create a custom dashboard so that I can monitor the KPIs most relevant to my team (by adding charts, tables, KPI cards, rearranging widgets, and applying filters such as time range).

- **Scenario:** Building a personalized dashboard

  Given the analyst opens the dashboard editor

  When they add, move, or remove widgets such as charts, tables, or KPI cards

  Then the system updates and saves the layout in real time

  And the configuration persists across sessions and devices

  And clear feedback is shown for each change

**User Story 11.2:** Select which data appears in dashboard

As a Marketing Analyst, I want to select which KPIs (e.g., conversion rate, ROI, engagement) appear on my dashboard so that I can tailor it to campaign goals.

- **Scenario:** Customizing KPI visibility

  Given the analyst opens the KPI selection panel

  When they add or remove KPIs like conversion rate or ROI

  Then the dashboard updates instantly with new metrics

  And selections are personal and do not affect other users

  And each KPI includes a description for clarity

**User Story 11.3:** Generate reports

As a Marketing Analyst, I want to generate an ROI report so that I can measure the financial effectiveness of marketing campaigns.

- **Scenario:** Producing ROI reports

  Given the analyst selects filters such as campaign or date range

  When they generate an ROI report

  Then the report displays clear sections and visuals

  And generation does not slow down dashboard updates

**User Story 11.4:** Export reports

As a Marketing Analyst, I want to export reports (e.g., PDF) so that I can share insights with other stakeholders.

- **Scenario:** Sharing reports externally

  Given the analyst views a report

  When they choose an export format and confirm

  Then the report is generated with correct values

  And the user receives confirmation when it's ready

And exports do not alter stored reports

**User Story 11.5:** Filter analytics by time

As a Marketing Analyst, I want to filter analytics by time period (e.g., last week, last month) so that I can identify trends over time.

- **Scenario:** Applying time-based filters

  Given the analyst chooses a predefined time range

  When the filter is applied

  Then all widgets update consistently

  And if there is no data, “No results” is displayed

  And time selection is intuitive with presets and a calendar picker

**User Story 11.6:** Track A/B test performance

As a Marketing Analyst, I want to track performance metrics for each A/B test so that I can determine the winning variation.

- **Scenario:** Comparing A/B test results

  Given the analyst selects an A/B test

  When metrics per variant are displayed

  Then the winning version is clearly highlighted

  And results feed into campaign reports and KPI dashboards

  And visualizations use distinct colors and labels

**User Story 11.7:** View social Media metrics

As a Marketing Analyst, I want to create charts with metrics for social media posts (likes, comments, shares, reach) so that I can evaluate content performance.

- **Scenario:** Monitoring post engagement

Given the analyst selects a social media post

When the system retrieves engagement data

Then likes, comments, shares, and reach are displayed clearly

And if API access fails, “Metrics unavailable” is shown

And visuals use clear icons and labels

**User Story 11.8:** Identify top-performing posts

As a Marketing Analyst, I want to identify top-performing and underperforming posts so that I can recommend which content types should be reused or expanded.

- **Scenario:** Ranking content performance

  Given engagement metrics are collected

  When the system ranks posts by performance

  Then top and underperforming posts are highlighted

  And rankings are shown with consistent colors and icons

## 2.5 Score

| Complexity | Very Easy | Easy | Medium | Hard | Very Hard |
|---|---|---|---|---|---|
| Story Points | 1 | 3 | 5 | 8 | 13 |

| Marketing Manager | |
|---|---|
| 1.1 User login | 5 |
| 1.2 User logout | 3 |
| 2.1 Capture and track metrics | 8 |
| 2.2 Group leads into segments | 5 |

| | |
|---|---|
| 2.3 Score leads based on actions | 5 |
| 3.1 Build automated marketing workflows | 13 |
| 3.2 Configure Triggers and Actions in Workflows | 8 |
| 3.3 Test and preview workflows | 8 |
| 3.4 Edit existing workflows | 5 |
| 4.1 Create and manage campaigns | 3 |
| 4.2 Manage campaign materials | 5 |
| 4.3 Run campaigns across multiple channels | 8 |
| 4.4 Track campaign results | 8 |
| 4.5 Export reports | 5 |
| 4.6 Track and update campaign status | 3 |
| 4.7 Create campaigns from reusable workflow templates | 5 |
| **Content Marketer** | |
| 5.1 Create and save Email Templates | 8 |
| 5.2 Edit Email Templates | 5 |
| 5.3 Personalize email templates | 8 |
| 5.4 Delete Email Templates | 2 |
| 5.5 Define A/B tests | 13 |
| 11.9 See Performance Metrics | 8 |
| 6.1 Create and preview Landing Pages | 5 |
| 6.2 Edit Landing Pages | 5 |
| 6.3 Delete Landing Pages | 2 |
| 7.1 Create Forms | 8 |
| 7.2 Edit Forms | 8 |
| 7.3 Delete Forms | 2 |
| 8.1 Schedule post for social media | 8 |
| 8.2 View Scheduled Posts | 5 |
| **Sales Representative** | |
| 9.1 Receive Notifications about leads | 8 |
| 9.2 View Leads email History | 8 |
| 9.3 Leads Webpage Visited | 8 |
| 9.4 Leads forms submitted | 8 |
| 9.5 Full leads timeline interactions | 8 |
| 9.6 Check Lead Score | 5 |

| | |
|---|---|
| 9.7 Sugested talking points based on the leads activity | 8 |
| 10.1 Sync Lead information with CRM | 8 |
| 10.2 Update Lead status in the CRM from our Platform | 8 |
| **Marketing Analyst** | |
| 11.1 Create Custom Dashboards | 5 |
| 11.2 Select which data appears in the dashboard | 13 |
| 11.3 Generate reports | 5 |
| 11.4 Export reports | 5 |
| 11.5 Filter analytics by time | 5 |
| 11.6 Track A/B test performance | 13 |
| 11.7 View social Media metrics | 8 |
| 11.8 Identify top-performing posts | 8 |

## 2.6 Requirements

### Functional Requirements (FR)

**Authentication**

FR1: The system shall allow login using email/username and password.

FR2: The system shall validate credentials and grant access when correct.

FR3: The system shall display a clear error message when credentials are invalid.

FR4: The system shall allow logout from any page.

FR5: The logout shall terminate the session and redirect the user to the login page.

**Lead Management**

FR6: The system shall capture leads via website forms.

FR7: The system shall store lead data in a centralized database.

FR8: The system shall allow searching and filtering of leads by defined criteria.

FR9: The system shall group leads into segments based on defined rules.

FR10: The system shall update segments when new leads are added.

FR11: The system shall assign scores to leads based on behavior and engagement.

FR12: The system shall maintain a history of each lead`s interactions.

**Campaigns**

FR13: The system shall support creation of multi-channel marketing campaigns.

FR14: The system shall allow creation of automated workflows triggered by lead actions.

FR15: The system shall allow creation of actions such as sending emails, updating lead scores, and notifying sales.

FR16: The system shall support A/B testing for campaigns.

FR17: The system shall allow personalized content based on lead/segment attributes.

FR18: The system shall generate performance reports for campaigns.

**Content**

FR19: The system shall provide customizable email templates.

FR20: The system shall allow creation, editing, and deleting of landing pages.

FR21: The system shall allow creation, editing, and deleting of forms for lead capture.

FR22: The system shall allow publishing and scheduling of social media posts.

**Sales Integration (CRM)**

FR23: The system shall synchronize leads and campaigns with CRM platforms.

FR24: The system shall notify the sales team automatically about qualified leads.

**Analytics**

FR25: The system shall provide interactive dashboards with real-time metrics.

FR26: The system shall calculate campaign KPIs.

FR27: The system shall generate detailed reports exportable in PDF/Excel.

### Non-Functional Requirements (NFR)

**Performance**

NFR1: The system shall support concurrent users without noticeable degradation.

**Security**

NFR2: User sessions shall automatically expire after a configurable inactivity period.

NFR3: Sensitive session data shall be deleted immediately after logout.

**Reliability**

NFR4: The system shall have a minimum availability of 99.99%.

**Usability**

NFR5: The user interface shall be intuitive and responsive across desktop.

NFR6: Analytics shall be available in both graphical and tabular formats.

**Maintainability**

NFR7: The codebase shall follow clean architecture principles.

NFR8: The system shall allow extension of features without impacting existing modules.

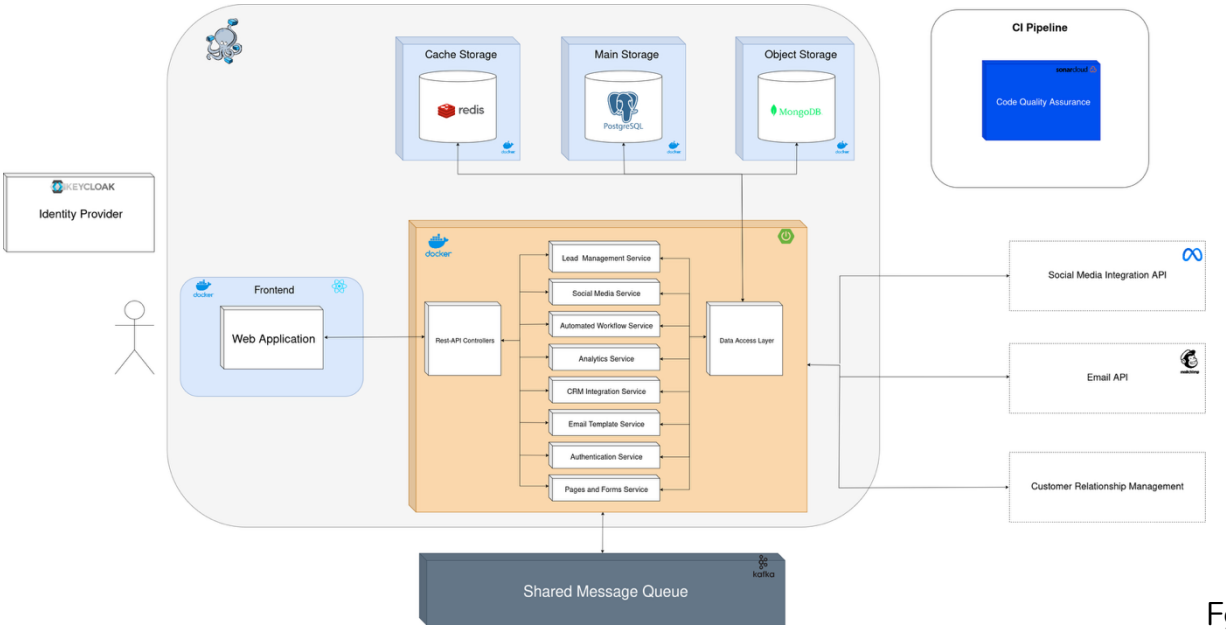
**Scalability**

NFR9: The system shall allow adding new modules without service interruption.

NFR10: The database shall support continuous growth without performance degradation.

# 3 Architecture notebook

## 3.1 Architetural view




Fg1. - MAP architecture

## 3.2 Module interactions

**Overview**

The system is designed as a modular and service-oriented platform, where each service exposes

REST APIs and communicates either synchronously through direct HTTP calls or asynchronously through the shared message queue (Kafka). The main actors include the **frontend (React web application)**, **backend microservices**, **identity provider (Keycloak)**, and **external integrations** (CRM, Social Media APIs, Email APIs). Data is persisted across different storage types: relational (PostgreSQL), object/document-based (MongoDB), and in-memory cache (Redis).

**Module Interactions**

**1. User Authentication and Authorization**

a. The Web Application uses **Keycloak** for authentication.
b. Keycloak handles user login and assigns **roles** (e.g., Marketing Manager, Content Marketer, Sales Rep, Analyst).
c. The frontend includes the Keycloak-issued token in API requests, and the system enforces access based on these roles. Analyst).

2. **Frontend Backend Communication**
   a. The frontend consumes the **REST API controllers**, which delegate requests to the appropriate backend service (Lead Management, Workflow, Analytics, etc.).
3. **Backend Services Interactions**
   a. **Lead Management Service** stores structured lead data in PostgreSQL and enriches lead profiles with interaction data from MongoDB.
   b. **Social Media Service** and **Email Template Service** integrate with external APIs (Meta API, Mailchimp) to publish content or send emails.
   c. **CRM Integration Service** synchronizes leads and campaign outcomes with external CRM systems (e.g., Salesforce, HubSpot).
   d. **Analytics Service** tracks Key Performance Indicators and grants visualization of the related data.
   e. **Automated Workflow Service** manages the design and execution of multi-step, automated marketing workflows
4. **Data Access Layer**
   a. Provides a unified abstraction over PostgreSQL (relational data), MongoDB (objects), and Redis (cache lookups).
   b. Ensures consistency across heterogeneous storage systems.